Momentum and Risk Analysis in US Large-Cap Technology Stocks

Quantitative Research Assignment

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Sector Chosen: US Large-Cap Technology
Stocks Analyzed: AAPL, MSFT, NVDA, AMZN, META
Time Period: January 2024 – May 2026

1. Objective

The aim of this analysis was to study how price behaves, how volatile it gets, and how returns relate across the big US technology equities. With historical market data, the report points out those visible market patterns, and then it suggests a fairly simple momentum driven trading strategy, plus a way to judge whether it actually works or not, over time.

Overall it concentrates on:

- finding patterns that really matter in the way stocks perform,
- getting a clearer view of sector risk traits,
- and turning the observations into a more usable investment idea.

2. Data Collection and Methodology

Selected Equities

The following large-cap technology companies were selected due to their strong market influence and high trading liquidity:

Company	Ticker
Apple	AAPL
Microsoft	MSFT

NVIDIA	NVDA
Amazon	AMZN
Meta Platforms	META

Data Source

Historical stock price data was collected using Yahoo Finance through Python's `yfinance` library.

Variables Used

The following variables were calculated:

Variable	Purpose
Adjusted Closing Price	Trend analysis
Daily Returns	Correlation and performance measurement
Trading Volume	Market activity analysis
20-Day Rolling Volatility	Risk measurement

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Yahoo Finance API (`yfinance`)

3. Key Findings and Observations

Finding 1 — NVIDIA Exhibited Strong Momentum and Significant Outperformance

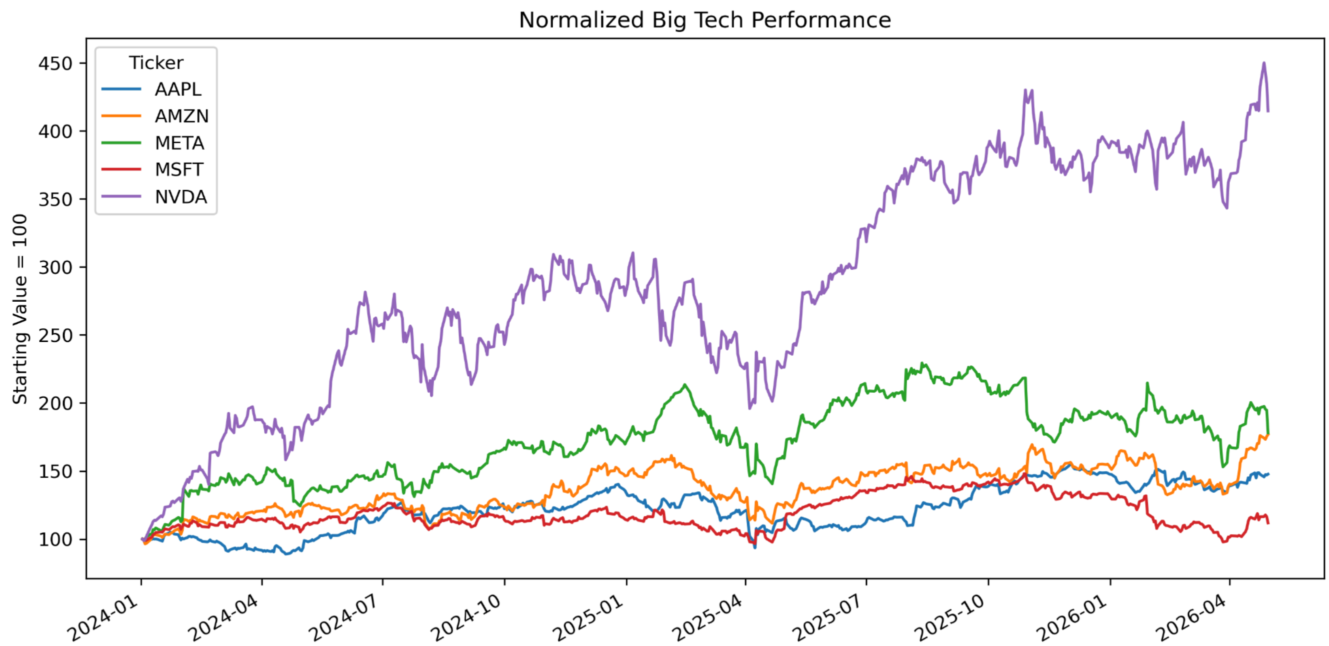


Figure 1: Normalized Performance of Selected Technology Stocks

The Figure 1 normalized performance chart shows that NVIDIA substantially outperformed the other selected technology equities during the sample period.

NVIDIA's normalized performance rose from a base value of 100 to about 450, which is way above peer firms like Apple and Microsoft.

Figure 1 seems to suggest, strong momentum persistence, sustained investor enthusiasm around AI-related businesses and positive earnings expectations. Also the stock's upward trend looked relatively persistent rather than something limited to one short-term rally only

Finding 2 — Higher Returns Were Accompanied by Higher Volatility

The rolling volatility chart demonstrated that NVIDIA consistently experienced higher volatility than the other selected equities.

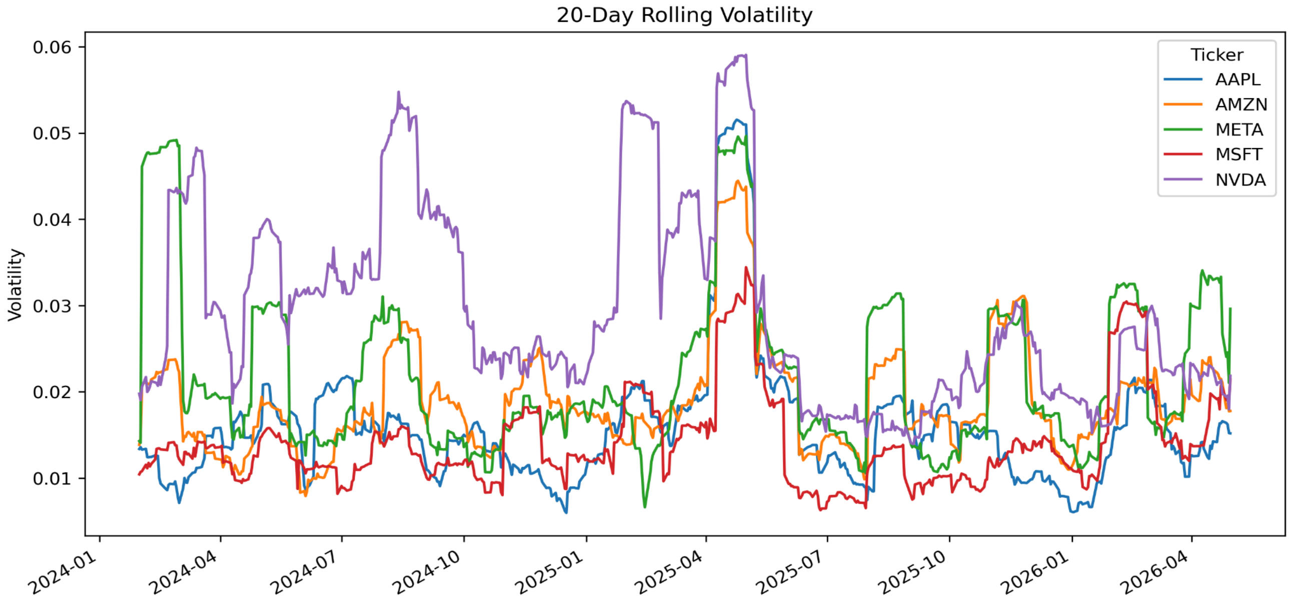


Figure 2: 20-Day Rolling Volatility

In Figure 2 Periods of sharp upward movement were often associated with spikes in rolling volatility, indicating that higher returns came alongside elevated risk exposure.

Interpretation

This reflects the classic risk-return tradeoff:

- investors demanded compensation for holding riskier growth-oriented equities,
- While AI-driven market sentiment amplified both returns and uncertainty.

Figure 2 finding also highlights that momentum-driven stocks may expose investors to larger drawdowns during periods of market correction.

Finding 3 — Technology Stocks Displayed Moderate-to-High Correlation

The correlation heatmap revealed positive correlations across all selected equities, with several stock pairs exhibiting moderately strong relationships.

For example:

- Amazon and Meta showed relatively high correlation,

- while most stock pairs remained positively related throughout the period.

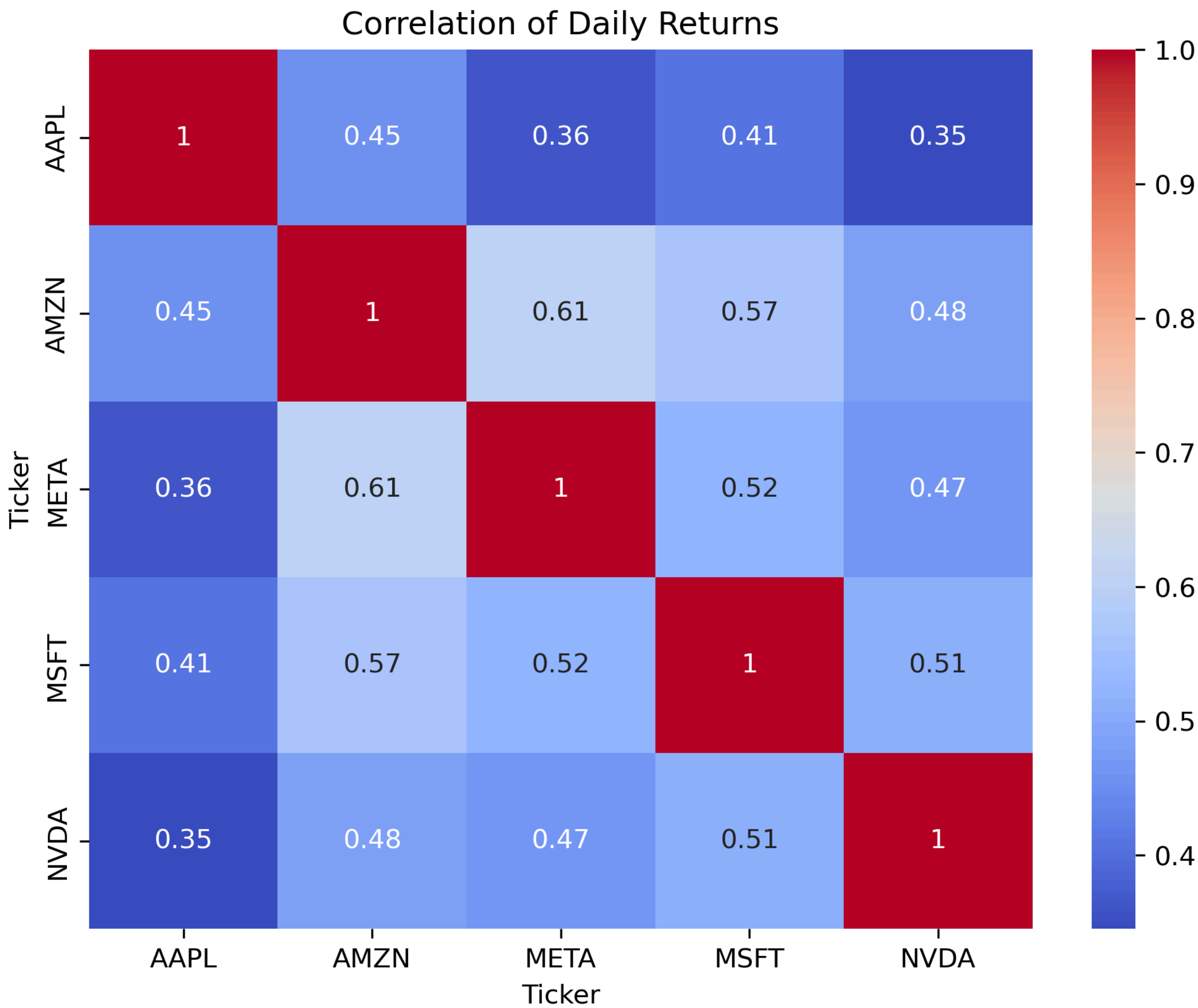


Figure 3: Correlation Heatmap of Daily Returns

Interpretation

Figure 3 suggests that, diversification benefits within the same sector might be only partially effective , and also that sector wide macroeconomic or market shocks can hit several firms at the same time. During periods of market stress, the correlations among technology equities can strengthen even more, which then raises the portfolio concentration risk .

4. Proposed Trading Idea

Momentum Rotation Strategy

Based on the observed persistence in stock outperformance, a simple momentum-based rotation strategy was proposed.

Strategy Rules

Every 20 trading days:

1. Calculate trailing 30-day returns for all five stocks.
2. Identify the best-performing stock.
3. Allocate capital to the top-performing stock.
4. Hold the position for the next 20 trading days.
5. Repeat the process at the next rebalance period.

Rationale Behind the Strategy

So basically the strategy comes from this, idea that the best technology equities, like NVIDIA especially, kept beating the market, over and over across different time windows. There are a few possible causes for this maybe not even one, more like several things at once. For example, institutional capital flows, or investor herding patterns , where people kind of follow what looks right. Also, there can be a delayed market reaction to growth expectations which can matter more than you think.

Overall the plan is trying to harness a short to medium term momentum persistence inside the technology sector, meaning when things start moving up they might keep doing it a bit longer than the crowd expects.

5. Backtesting Framework

To evaluate the usefulness of the proposed strategy, a historical backtest would be conducted using the same dataset.

Backtesting Steps

1. Compute daily returns for all stocks.
2. Rank equities using trailing 30-day returns.
3. Simulate portfolio allocation to the highest-ranked stock.

4. Rebalance every 20 trading days.
5. Compare results against a benchmark portfolio.

Benchmark

The strategy would be compared against:

- an equal-weight portfolio containing all five equities.

Performance Metrics

The following metrics would be evaluated:

Metric	Purpose
Cumulative Return	Total profitability
Volatility	Risk measurement
Sharpe Ratio	Risk-adjusted return
Maximum Drawdown	Downside risk
Win Rate	Strategy consistency

Additional Assumptions

The backtest should also account for:

- transaction costs,
- rebalancing frequency,
- and slippage assumptions.

Including these assumptions would make the results more realistic.

6. Risks and Limitations

Although the strategy appears conceptually reasonable, several important limitations exist.

1. Overfitting Risk

The observed momentum effect may be heavily influenced by the strong AI-driven market environment during the sample period.

The strategy may not perform similarly under different market conditions.

2. Limited Sample Size

The analysis covers approximately two years of market data, which may not fully capture:

- recessionary periods,
- bear markets,
- or multiple economic cycles.

3. Momentum Reversal Risk

Momentum strategies can perform poorly during sudden market reversals when previously outperforming stocks rapidly decline.

4. Concentration Risk

The strategy allocates capital to only one stock at a time, exposing investors to substantial company-specific risk.

5. Transaction Costs

Frequent portfolio rebalancing may reduce net returns, especially in real-world implementation.

7. Conclusion

This analysis identified several important characteristics within large-cap technology equities between 2024 and 2026.

Key findings included:

- strong momentum persistence in NVIDIA,

- elevated volatility among high-growth equities,
- and moderate-to-high correlations across the sector.

Based on these observations, a simple momentum rotation strategy was proposed and a framework for testing its effectiveness was outlined.

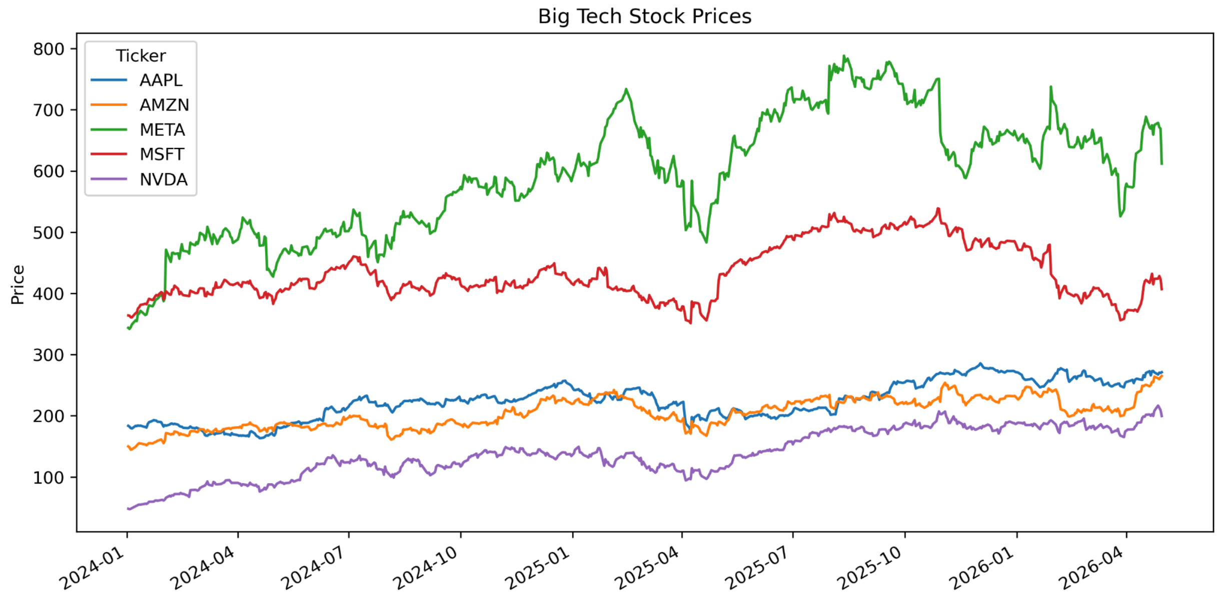
Although the strategy may benefit from momentum persistence during bullish periods, it remains exposed to volatility, concentration risk, and changing market conditions.

Appendix

Appendix A — Python Code for Data Collection

```
stocks = ["AAPL", "MSFT", "NVDA", "AMZN", "META"]

data = yf.download(
    stocks,
    start="2024-01-01",
    end="2026-05-01"
)

close = data["Close"]

returns = close.pct_change()

volatility = returns.rolling(20).std()
```

Appendix B — Price Trend Chart



Appendix C — Backtesting Logic

Simple pseudocode like:

1. Calculate 30-day returns
2. Rank stocks
3. Select top performer
4. Hold for 20 days
5. Rebalance

AI Usage Disclosure

AI tools were used to:

- assist with structuring the report,
- refine explanations,

- and improve clarity of financial interpretation.

All stock selection, dataset generation, visualization, observations, and analytical conclusions were reviewed and validated manually.